

CV – YASSINE HAMOUDI

OVERVIEW

- First/Last name: Yassine HAMOUDI
- Address: LaBRI, Bureau 375, 351 cours de la Libération, 33405 Talence, FRANCE
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Research Activities

My primary research interest is *quantum computing*. I study the impact of quantum mechanics on algorithm design and computational complexity theory. I have contributed to new quantum *algorithms* in diverse areas such as optimization, statistical analysis, streaming data, graph problems and cryptanalysis. I am also interested in new algorithmic and *lower-bound* methods for understanding the interplay between the computational resources available to quantum computers, with a particular interest in query complexity.

EMPLOYMENT AND EDUCATION

CNRS junior researcher (“chargé de recherche”)

- LaBRI, Université de Bordeaux, France – October 2023–present.
- Head of the “Quantum Information and Computing” group: <https://quantique.labri.fr/>.

Postdoctoral researcher

- University of California, Berkeley, United States – September 2021 to August 2023.

Doctoral degree

- IRIF, Université Paris Cité, France – September 2017 to June 2021.

Bachelor’s and Master’s Degrees in Computer Science

- Ecole Normale Supérieure de Lyon, France – September 2013 to August 2017.

PUBLICATIONS

Peer-reviewed conferences

- “Dequantization Barriers for Guided Stoquastic Hamiltonians”,
Contributed talk at the 21st Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC), 2026.
Yassine Hamoudi, Yvan Le Borgne and Shrinidhi Teganahally Sridhara.
ArXiv: 2602.23183.
- “Quantum Speedup of Non-Linear Monte Carlo Problems”,
Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS), 2025. **“Spotlight Talk”**.
Jose Blanchet, Yassine Hamoudi, Mario Szegedy and Guanyang Wang.
DOI: 10.52202/085713-0631. ArXiv: 2502.05094.
- “The NISQ Complexity of Collision Finding”,
Proceedings of the 43rd International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT), 2024.
Yassine Hamoudi, Qipeng Liu and Makrand Sinha.
DOI: 10.1007/978-3-031-58737-5_1. ArXiv: 2211.12954.
- “A Sublinear-Time Quantum Algorithm for Approximating Partition Functions”,
Contributed talk at the 26th Conference on Quantum Information Processing (QIP), 2023.
Proceedings of the 34th Symposium on Discrete Algorithms (SODA), 2023.
Arjan Cornelissen and Yassine Hamoudi.
DOI: 10.1137/1.9781611977554.ch46. ArXiv: 2207.08643.
- “Classical and Quantum Algorithms for Variants of Subset-Sum via Dynamic Programming”,
Proceedings of the 30th European Symposium on Algorithms (ESA), 2022.
Jonathan Allcock, Yassine Hamoudi, Antoine Joux, Felix Klingelhöfer and Miklos Santha.
DOI: 10.4230/lipics.esa.2022.6. ArXiv: 2111.07059.
- “Near-Optimal Quantum Algorithms for Multivariate Mean Estimation”,
Contributed talk at the 25th Conference on Quantum Information Processing (QIP), 2022.
Proceedings of the 54th Symposium on Theory of Computing (STOC), 2022.
Arjan Cornelissen, Yassine Hamoudi and Sofiene Jerbi.
DOI: 10.1145/3519935.3520045. ArXiv: 2111.09787.
- “Quantum Sub-Gaussian Mean Estimator”,
Proceedings of the 29th European Symposium on Algorithms (ESA), 2021.
Yassine Hamoudi.
DOI: 10.4230/lipics.esa.2021.50. ArXiv: 2108.12172.
- “Quantum Time-Space Tradeoff for Finding Multiple Collision Pairs”,
Proceedings of the 16th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC), 2021. **“Outstanding Paper Award”**.
Yassine Hamoudi and Frédéric Magniez.
DOI: 10.4230/lipics.tqc.2021.1. ArXiv: 2002.08944.

- “Quantum Chebyshev’s Inequality and Applications”,
Proceedings of the 46th International Colloquium on Automata, Languages, and Programming (ICALP), 2019.
Yassine Hamoudi and Frédéric Magniez.
DOI: 10.4230/lipics.icalp.2019.69. ArXiv: 1807.06456.
- “Simultaneous Multiparty Communication Complexity of Composed Functions”,
Proceedings of the 43rd International Symposium on Mathematical Foundations of Computer Science (MFCS), 2018.
Yassine Hamoudi.
DOI: 10.4230/lipics.mfcs.2018.14. ArXiv: 1710.01969.

Peer-reviewed journals

- “Preparing Many Copies of a Quantum State in the Black-Box Model”,
Physical Review A, 2022.
Yassine Hamoudi.
DOI: 10.1103/physreva.105.062440. ArXiv: 2207.11014.
- “Quantum Algorithms for Hedging and the Learning of Ising Models”,
Physical Review A, 2021.
Patrick Rebentrost, Yassine Hamoudi, Maharshi Ray, Miklos Santha, Xin Wang and Siyi Yang.
DOI: 10.1103/physreva.103.012418. ArXiv: 2002.06003.
- “Quantum and Classical Algorithms for Approximate Submodular Function Minimization”,
Quantum Information & Computation, 2019.
Yassine Hamoudi, Patrick Rebentrost, Ansis Rosmanis and Miklos Santha.
DOI: 10.26421/qic19.15-16-5. ArXiv: 1907.05378.

Preprints

- “A Brief Introduction to Quantum Query Complexity”,
In submission, 2025.
Yassine Hamoudi.
ArXiv: 2508.08852.

SCIENTIFIC RESPONSIBILITIES

Service

- Member of the Executive Committee of the Naquidis+ initiative (regional quantum center of Nouvelle-Aquitaine).
- Head of the “Quantum Information and Computing” group at LaBRI: <https://quantique.labri.fr/>.

Grants

- Coordinator of the project “Maisons du Quantique - Nouvelle-Aquitaine” (CEA/GENCI grant of €1.2 million): <https://quantique.labri.fr/hybquant/> – 2024 to 2028.

- Participant to the project EPiQ (PEPR integrated project): <https://project.inria.fr/epiq/> – 2022 to 2029.

Advising

- Karim El Houdaigui (Research engineer) – April 2026 to present.
- Joseph Cunningham (Postdoctoral researcher) – November 2025 to present.
- Vishvajeet Nagargoje (Postdoctoral researcher) – October 2025 to present.
- Shrinidhi Teganahally Sridhara (PhD student), co-advised with Adrian Tanasa – October 2024 to present.
- Ananya Rajas Ranade (Undergraduate intern) – May to July 2025.
- Rishav Gupta (Undergraduate intern), co-advised with Adrian Tanasa – May to July 2024.

Examination Committees

- Member of the PhD Thesis Committee of Xiaoyu Sun, Aix-Marseille Université, France, 2026. “Quantum perceptron algorithms and machine learning techniques for quantum separability”.
- Member of the PhD Thesis Committee of Joseph Cunningham, Université Libre de Bruxelles, Belgium, 2025. “Alternative adiabatic quantum dynamics with algorithmic applications”.

Program committees and reviewer

- Program committee member: QTML 2022, TQC 2023, TQC 2024, QCTiP 2025, QIP 2026, ESA 2026.
- Conference reviewer: ESA 23,24; FCT 23; FOCS 20,21; FSTTCS 20,21; ICALP 19,21,22,23,25, 26; ITCS 20,25; MFCS 18,21,22,23,24; QCTiP 26; QIP 21,22,23,24,25; SAC 21; SoCG 22; SODA 21,22,23,24,25; STACS 21,23,24,25; STOC 23,24,25,26; TQC 22,25,26.
- Journal reviewer: ACM Transactions on Quantum Computing, Algorithmica, Discrete Mathematics & Theoretical Computer Science, IEEE Transactions on Computers, International Journal of Quantum Information, Mathematics of Operations Research, Quantum, Quantum Information & Computation, SIAM Journal on Computing.
- Scientific expert for the Belgian FNRS PhD Fellowship, 2024 & 2026.
- Grader for the written entrance examinations to École Polytechnique and ENS, 2024 to 2026.

Organization of scientific events

- Workshop “2nd Basque Country - Bordeaux Quantum Algorithms Meeting” (Bayonne, France): <https://quantique.labri.fr/bayonne2026/> – March 2026.
- CNRS GT IQ meeting “Journées Informatique Quantique” (LaBRI, France): <https://quantique.labri.fr/jiq2026/> – January 2026.
- Workshop “Basque Country - Bordeaux Quantum Algorithms Meeting” (Bayonne, France): <https://quantique.labri.fr/bayonne2025/> – April 2025.

- Workshop “Quantum Algorithms for Massive Data” (LaBRI, France): https://dept-info.labri.fr/~gavoille/qudata/meeting_bordeaux_jan2024.html – January 2024.
- Group seminar “Quantum Information & Computation” (LaBRI, France): <https://quantique.labri.fr/seminar/> – 2024 to present.
- Group seminar “Algorithms and Complexity” (IRIF, France): <https://www.irif.fr/en/seminaires/algocomp/> – 2017 to 2021.

TALKS

Invited talks and courses

- Lecturer at the *CEMRACS/CIRM Summer School on Quantum Computing*, Marseille, France, July 2025. “Optimization Problems on Quantum Computers”.
- Lecturer at the *UPV/EHU Summer Course on Quantum Technologies*, San Sebastian, Spain, September 2024. “Introduction to Quantum Computing”.
- Invited speaker at the *American Physical Society (APS) March Meeting*, Las Vegas, United States, March 2023. “Quantum Speedups for Computing Expectation Values and Partition Functions”.
- Lecturer at the *IAS/Park City Mathematics Institute (PCMI) Graduate Summer School on Quantum Computation*, United States, July 2023. “Quantum Query Complexity”.

Scientific outreach

- Opening talk on quantum algorithms at the *Journée Informatique Quantique Toulouse*, TSM, France, 29 June 2026.
- Outreach talk on “Les algorithmes à l’ère du calcul quantique” at the *Conférence sur les nouveaux paradigmes de calcul*, Siège du CNRS, France, 8 October 2025.
- Inaugural presentation of the “Maison du Quantique - Nouvelle-Aquitaine” at the *Journée Quantique en Nouvelle-Aquitaine*, Institut d’Optique d’Aquitaine, France, 8 July 2025.

Conference talks

- *International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT)*, Zurich, Switzerland, 29 May 2024.
- *International Conference on Quantum Information Processing (QIP)*, Ghent University, Belgium, 8 February 2023. Video: <https://www.youtube.com/watch?v=BI3hi3p-jsg>.
- *European Symposium on Algorithms (ESA)*, Hasso Plattner Institut, Germany, 7 September 2022 (virtual). Video: <https://youtu.be/EI02dxqVQ8U>.
- *Symposium on Theory of Computing (STOC)*, Rome, Italy, 20 June 2022. Video: <https://youtu.be/D00VreIL9so>.
- *European Symposium on Algorithms (ESA)*, Instituto Superior Técnico, Portugal, 7 September 2021 (virtual). Video: <https://youtu.be/dlvme3gUJcY>.

- *Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC)*, University of Latvia, 6 July 2021 (virtual). Video: <https://youtu.be/eXvznaL2d-M>.
- *International Colloquium on Automata, Languages and Programming (ICALP)*, University of Patras, Greece, 9 July 2019.
- *International Symposium on Mathematical Foundations of Computer Science (MFCS)*, University of Liverpool, United Kingdom, 27 August 2018.

Seminars

- *QInfo Seminar*, ENS Lyon, France, 11 December 2025.
- *Séminaire du Département Informatique*, ENS Lyon, 10 December 2025.
- *Numerics Seminar*, Université de Bordeaux, France, 16 October 2024 & 2025.
- *COSMIQ Seminar*, INRIA Paris, France, 30 January 2023.
- *Quantum Information Seminar*, LIP6, France, 8 December 2022.
- *GT Info-Quantique*, LaBRI, France, 6 December 2022.
- *CAPP Seminar*, LIG, France, 30 November 2022.
- *QInfo Seminar*, École normale supérieure de Lyon, France, 28 November 2022.
- *Stanford Theory Lunch*, Stanford University, United States, 9 June 2022.
- *Quantum Brainstorming Session*, Simons Institute for the Theory of Computing, United States, 19 November 2021.
- *GT Info-Quantique*, LaBRI, France, 9 November 2021 (virtual).
- *QuSoft Seminar*, CWI, Netherlands, 8 October 2021 (virtual).
- *Bristol QIT Seminar*, University of Bristol, United Kingdom, 4 March 2020.
- *Centre for Quantum Technologies Seminar*, CQT, NUS, Singapore, 6 March 2019.

Workshops

- *Basque Country - Bordeaux Quantum Algorithms Meeting*, Bayonne, France, 9 April 2025.
- *Quantum Algorithms for Massive Data (QUDATA)*, LaBRI, France, 26 January 2024.
- *Quantum Summer Cluster Workshop*, Simons Institute, United States, 11 July 2023.
- *CIQC Annual Meeting*, UC Berkeley, United States, 12 June 2023.
- *Quantum Algorithms for Massive Data (QUDATA)*, LaBRI, France, 26 January 2024.
- *Quantum Algorithms and Complexity*, IQC, University of Waterloo, Canada, 25 October 2019.
- *Quantum Algorithms and Applications (QUANTALGO)*, CWI, Netherlands, 18 September 2019.
- *Quantum Algorithms for Massive Data (QUDATA)*, IRIF, France, 10 January 2019.

- *Journées Informatique Quantique* (JIQ), Loria, France, 8 November 2018.
- *Quantum Algorithms and Applications* (QUANTALGO), IRIF, France, 25 September 2018.

Invited visits

- Simons Institute, United States, April–May 2024.
- Columbia University, United States, October 2022.
- University of Bristol, United Kingdom, March 2020.
- University of Waterloo, Canada, October 2019.
- Centre for Quantum Technologies, Singapore, April–June 2017, February–June 2019.
- Carnegie Mellon University, United States, January–June 2016.